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ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

CHEMISTRY
PAPER 1

9189/1

JUNE 2012 SESSION

2 hours

Additional materials:
Answer paper
Data Booklet
Mathematical tables and/or electronic calculator

TIME: 2 hours

INSTRUCTIONS TO CANDIDATES

Write your name, Centre number and candidate number in the spaces provided on the answer paper/answer booklet.

Answer **six** questions.

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This question paper consists of 8 printed pages.

[Turn over

Section A

Answer at least two questions from this section.

1 (a) (i) Using iron as an example, explain the term *standard electrode potential.*

(ii) Draw a well-labelled diagram to show how the standard electrode potential of iron can be measured in a laboratory.

(iii) Suggest why the standard electrode potential of calcium cannot be measured directly by the method in (ii).

[5]

(b) [Use of the Data Booklet is relevant to this question].

Acidified solutions of 1.0 $moldm^{-3}$ potassium manganate (VII), $KMnO_4$; manganese (II) sulphate, $MnSO_4$; potassium dichromate (VI), $K_2Cr_2O_7$ and chromium (III) sulphate, $Cr_2(SO_4)_3$ were mixed and allowed to come to equilibrium.

(i) Construct an equation for the overall reaction and calculate the E^θ cell.

(ii) State which ion was oxidised and which was reduced

[4]

(c) Solid oxide fuel cells that use oxygen and butane as fuel are in a developmental stage. They contain composite metal/metal oxide electrodes and an unusual solid metal oxide electrolyte. The cathode reaction equation is:

$$O_2 + 4e^- \rightarrow 2O^{2-}$$

(i) Construct an anode reaction equation given that the anode reaction involves a reaction of butane with the oxide ions to produce carbon dioxide and water.

(ii) Write a balanced equation for the cell reaction.

(iii) Suggest one possible disadvantage of this cell compared to the hydrogen/oxygen fuel cell.

[3]

[Total: 12]

2 (a) Reforming petroleum involves cracking larger hydrocarbon molecules into smaller more useful species under suitable conditions. For instance, gaseous butane can be cracked to ethane and ethene, the process reaching equilibrium.

(i) Given that the equation for the cracking of butane is

$$C_4H_{10(g)} \rightleftharpoons C_2H_{6(g)} + C_2H_{4(g)},$$

write the equilibrium expression, K_p for the cracking of butane.

(ii) A sample of butane, having a pressure of 50 atm, is heated at 600°C in a closed container at constant volume.

Given that the equilibrium partial pressures for each of the products is 19.2 atm, calculate

1. the equilibrium constant,

2. the percentage of butane converted to ethane and ethene,

3. the total final pressure.

(iii) Explain the effect of increasing volume on the equilibrium in (i). [8]

(b) Ethanoic acid dimerises in inert solvents such as benzene. The numerical value for the equilibrium constant, K_c for this dimerisation is 1.51×10^2 in benzene but only 3.70×10^{-2} in water.

(i) Draw the structure of the dimmer.

(ii) Give two reasons for the difference in the two K_c values.

(iii) Predict any differences in pH expected between the two solutions.

[4]
[Total: 12]

[Turn over

3 (a) Define the terms

$xA + yB \rightarrow$ products

Rate $= k[A]^m[B]^n$

(i) *rate equation* and

(ii) *order of reaction.*

[2]

(b) Phenylamine reacts with nitrous acid in the presence of concentrated hydrochloric acid between 0 and 5°C to produce benzenediazonium chloride. If the temperature of this solution rises, benzenediazonium chloride decomposes in aqueous solution to give phenol and nitrogen.

Given that the equation for the decomposition of benzenediazonium chloride in aqueous solution at elevated temperatures is

$$C_6H_5N_2Cl + H_2O \rightarrow C_6H_5OH + N_2 + HCl,$$

suggest two different experimental methods for studying the kinetics of this reaction.

[2]

(c) At 60°C the concentration of benzenediazonium chloride was found to change with time as shown in **Table 1**.

Table 1

[benzenediazonium chloride]/ moldm^{-3}	0.160	0.1160	0.080	0.064	0.048.	0.034	0.016
time / minutes	0	2	4	6	8	10	15

(i) Plot a graph of these results and use it to deduce the order of reaction with respect to benzenediazonium chloride.

(ii) Write the rate equation for the reaction and calculate the rate constant at 60°C.

[8]

[Total: 12]

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

CHEMISTRY **9189/1**

PAPER 1

JUNE 2013 SESSION 2 hours

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Section A

Answer at least two questions from this section.

1 (a) Draw labelled diagrams to show crystalline structures of:

(i) magnesium oxide,

(ii) ice. [2]

(b) Using the diagrams in (a) where necessary, explain the following observations:

(i) magnesium oxide is partially soluble in water

(ii) melting point of ice is less than that of magnesium oxide

(iii) ice floats on water

(iv) lattice energy of magnesium oxide is greater than that of sodium chloride which has a similar crystalline structure [4]

(c) (i) Construct a Born-Haber cycle for the formation of magnesium oxide from its elements.

(ii) Use the cycle to calculate the enthalpy change of formation of magnesium oxide. Use the following data as well as relevant data from the Data Booklet.

$\Delta H_a(Mg) = +150\ kJmol^{-1}$

$\Delta H_{latt}(MgO) = -3933\ kJmol^{-1}$

First electron affinity of oxygen = $-142\ kJmol^{-1}$

Second electron affinity of oxygen = $+775\ kJmol^{-1}$

(iii) Explain why the second electron affinity of oxygen is endothermic. [6]

[Total: 12]

2 (a) Define a *Bronsted-Lowry acid.* [1]

(b) Explain the following observations:

(i) $CCl_3CO_2H_{(aq)}$ is a stronger acid than $CH_3CO_2H_{(aq)}$

(ii) $[Al(H_2O)_6]^{3+}{}_{(aq)}$ forms a stronger acid than $[Mg(H_2O)_6]^{2+}$ [3]

(c) A 15.0 cm^3 sample of 0.1 $moldm^{-3}$ aqueous sodium hydroxide was added to 25.0 cm^3 of 0.1 $moldm^{-3}$ ethanoic acid of pk_a value 4.76.

(i) Explain, with the aid of equations, why the resulting solution can act as a buffer solution.

(ii) Calculate the pH of the resulting solution.

(iii) Draw a sketch of the titration curve obtained when 50.0 cm^3 of 0.1 $moldm^{-3}$ ethanoic acid is titrated with 25.0 cm^3 of 0.2 $moldm^{-3}$ sodium hydroxide. [8]

[Total: 12]

3 **(a)** Define the term *half life*. [1]

(b) Hydrolysis of methylethanoate, CH_3COOCH_3, is catalysed by hydrogen ions, H^+. The following results were obtained when the concentration of the ester was measured as a function of time, *t*, at different hydrogen ion concentrations.

At $[H^+] = 0.3$ $moldm^{-3}$:

[ester]/ $moldm^{-3}$	1.00	0.64	0.41	0.26
t/s	0.00	1500	3000	4500

At $[H^+] = 0.6$ $moldm^{-3}$:

[ester]/ $moldm^{-3}$	1.00	0.55	0.30	0.17
t/s	0.00	1000	2000	3000

(i) Write the equation for the acid hydrolysis of methylethanoate.

(ii) Using the same axes, plot graphs of [ester] against time at the different concentrations of hydrogen ions.

(iii) Using the initial rates-method, determine from the graphs, the order of the reaction with respect to

1. [ester],

2. [hydrogen ions].

(iv) Write the rate equation for the reaction. [11]

[Total: 12]

66

ZIMSEC
For Performance Measurement

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

CHEMISTRY

PAPER 1

9189/1

JUNE 2014 SESSION

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- Graph paper
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TIME: 2 hours

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22

Section A

*Answer at least **two** questions from this section.*

1 **(a)** **(i)** Write the electronic configuration of

1. calcium,

2. chlorine.

(ii) Draw a dot-and-cross diagram to illustrate the type of bond formed between calcium and chlorine.

[3]

(b) **(i)** Define the term *enthalpy change of hydration.*

(ii) Write an equation for the enthalpy change of hydration of the

1. calcium ion,

2. chloride ion.

(iii) Given that:

lattice energy of calcium chloride = $-2\ 258$ $kJmol^{-1}$
enthalpy change of hydration of calcium ions = $-1\ 650$ $kJmol^{-1}$
enthalpy change of solution of calcium chloride = -120 $kJmol^{-1}$

1. construct an energy cycle for these enthalpies.

2. use the cycle to calculate the enthalpy change of hydration of the chloride ion.

[6]

(c) The enthalpy change of hydration of magnesium ion is $-1\ 921$ $kJmol^{-1}$.

(i) Use your value in **(b)(iii)** to calculate the total enthalpy change of hydration of the ions in magnesium chloride.

(ii) Comment on the difference between the enthalpy of hydration of the magnesium and calcium ions.

[3]
[Total: 12]

2 **(a)** With the aid of a labelled diagram, describe the electrolysis of brine using a diaphragm cell. [7]

(b) By reference to relevant E^θ values from the Data Booklet,

(i) explain the disadvantage of using a silver anode,

(ii) justify why electricity must be supplied for the electrolysis of brine.

[5]

[Total: 12]

3 **(a)** The following equation represents an equilibrium reaction:

$$\underset{\text{colourless}}{N_2O_{4(g)}} \rightleftharpoons \underset{\text{brown}}{2NO_{2(g)}} \quad \Delta H^\theta = +58\ kJmol^{-1}$$

(i) State LeChatelier's principle.

(ii) Describe and explain the colour change(s) if any, that might take place when

1. a mixture of $NO_{2(g)}$ and $N_2O_{4(g)}$ is compressed at constant temperature,

2. a mixture of $NO_{2(g)}$ and $N_2O_{4(g)}$ is heated at constant pressure.

[5]

(b) In an experiment to determine the equilibrium constant for the reaction in **(a)** at 200°C, the initial concentrations of N_2O_4 and NO_2 were 0.075 moldm^{-3} and 0.0250 moldm^{-3} respectively. If the equilibrium concentration of N_2O_4 was 0.00275 moldm^{-3},

(i) calculate the equilibrium concentration of NO_2,

(ii) determine the K_c value of the N_2O_4/NO_2 system at 200°C.

[4]

(c) NO_2 can catalyse the conversion of atmospheric sulphur dioxide to sulphur trioxide.

(i) Write equations to illustrate this conversion.

(ii) Explain the type of catalysis exhibited by NO_2 in this conversion.

[3]

[Total: 12]

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NOVEMBER 2014 SESSION 2 hours

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Mathematical tables and/or electronic calculator
Graph paper

TIME: 2 hours

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Section A

*Answer at least **two** questions from this section.*

1 **(a)** Define the term *relative isotopic mass*. [1]

(b) An analysis of the composition of a newly discovered element, Q, showed the following results

Isotope	Relative abundance (%)
^{20}Q	90.91
^{21}Q	0.16
^{22}Q	8.93

On the mass spectrum of Q, the peak due to isotope ^{21}Q had a peak of 0.20 cm.

(i) Deduce the heights of the other two peaks.

(ii) Draw a sketch of the mass spectrum of Q in terms of these heights.

(iii) Calculate the relative atomic mass of Q.

[4]

(c) A 1.50 g sample of anhydrous iron (II) ethane-dioate, FeC_2O_4, was dissolved in 100.00 cm^3 of water. A 25.00 cm^3 sample of the solution required 15.00 cm^3 of $KMnO_{4(aq)}$ for a complete reaction titration.

(i) Use the following half equations to write the overall equation for the oxidation of FeC_2O_4 by $KMnO_4$.

$$2CO_2 + 2e^- \rightleftharpoons C_2O_4^{2-}$$

$$Fe^{3+} + e^- \rightleftharpoons Fe^{2+}$$

$$MnO_4^- + 8H^+ + 5e^- \rightleftharpoons Mn^{2+} + 4H_2O$$

(ii) Calculate the concentration of $KMnO_{4(aq)}$. [7]

[Total: 12]

2 **(a)** Define an *acid-base indicator*. [1]

(b) If an indicator can be conveniently represented as HIn and its equilibrium dissociation constant as KIn, write

(i) the equation for the dissociation of HIn in aqueous solution,

(ii) an expression for KIn.

[2]

(c) Congo red, an acid base indicator, is blue in acidic conditions and red in alkaline conditions. It has a KIn value of 1.2×10^{-4} moldm^{-3}.

(i) Given that colour change begins when $\left[In^{-}\right] = \left[HIn\right]$, calculate the pH at which Congo red begins to change colour.

(ii) State the acid-base titrations for which Congo red would be most suitable.

(iii) Calculate the pH of a 0.2 moldm^{-3} solution of Congo red.

(iv) If an Indicator contributes to the pH of a solution, suggest a reason why the pH of a titration mixture is not affected by the Indicator used. [5]

(d) The pH changes during the titration of acid **HA** against base **HB** are shown in **Table 1**.

Table 1

Volume of base **HB**/cm^3	0	5	10	15	20	22	25	30
pH	0.8	1.0	1.6	2.4	3.6	8.0	9.0	9.6

(i) Plot a graph of pH against volume of base **HB**.

(ii) Use your graph to deduce the pH at end point.

[4]
[Total: 12]

[Turn over

3 (a) (i) Explain the *phrase mechanism of a reaction.*

(ii) Suggest **one** reason why the study of reaction kinetics is of importance to chemists.

[2]

(b) The reaction between two gases **A** and **B** is represented by the equation:

$$2A + B \rightarrow A_2B$$

Two experiments were carried out in which the change in the partial pressure of each gas with time was monitored.

The results obtained are shown in the tables:

Experiment 1: Change in partial pressure of **A** with time

Time/s	0	600	1 200	1 800	2 400
Partial pressure of A/atm	0.500	0.250	0.170	0.125	0.080

Experiment 2: Change in Partial pressure of **B** with time

Time/s	0	600	1 200	1 800	2 400
Partial pressure of B/atm	0.450	0.375	0.300	0.225	0.150

(i) On the same axes plot graphs of change in partial pressure against time for each reactant. Label each graph.

(ii) Use your graphs to deduce the order of reaction with respect to:

1. **A**

2. **B**

(iii) Suggest a suitable mechanism for the reaction.

[10]
[Total: 12]

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

CHEMISTRY

PAPER 1 Theory

9189/1

JUNE 2015 SESSION

2 hours

Additional materials:
- Answer paper
- Data Booklet
- Mathematical tables and/or electronic calculator

TIME: 2 hours

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[Turn over

Section A

*Answer at least **two** questions from this section.*

1 (a) (i) Define the phrase *first ionization energy* of an element.

(ii) Write an equation for the first ionization energy of chlorine. [2]

(b) (i) **Table 1** shows the first ionisation energies of four elements.

Table 1

element	1^{st} ionization energy/kJ mol^{-1}
magnesium	736
aluminium	577
phosphorus	1 060
sulphur	1 000

Explain why

1. magnesium has a higher first ionisation energy than aluminium,
2. sulphur has a lower first ionisation energy than phosphorus.

[4]

(ii) The successive ionization energies of an element, M, in kJ mol^{-1} are:

1^{st}	2^{nd}	3^{rd}	4^{th}	5^{th}
740	1 500	7 700	10 500	13 600

State, with reasons, the group of the periodic table to which M belongs. [3]

(c) State and explain the variation of electrical conductivity of Period (III) elements, Na to Ar. [3]

[Total:12]

2 (a) **Fig. 2.1** shows the structural formulae of isomeric hydrocarbons, P and Q.

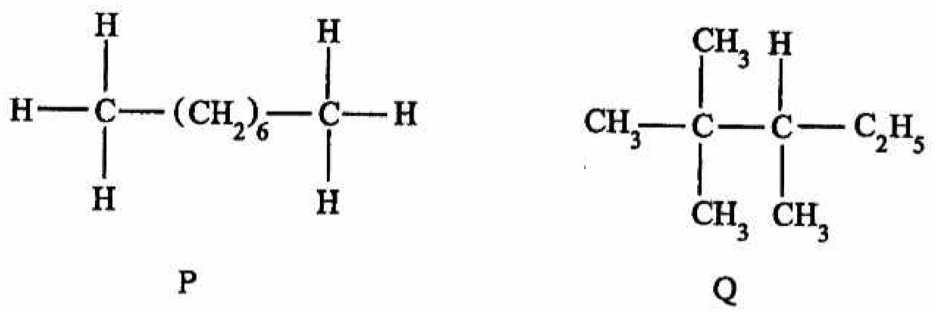

Fig. 2.1

(i) Identify, with reasons, the hydrocarbon with a higher boiling point.

(ii) Calculate the percentage of carbon by mass in 1 mole of P.

[4]

(b) **Fig. 2.2** shows the trend in the boiling points of the hydrides of Group (VI) elements, oxygen to tellurium.

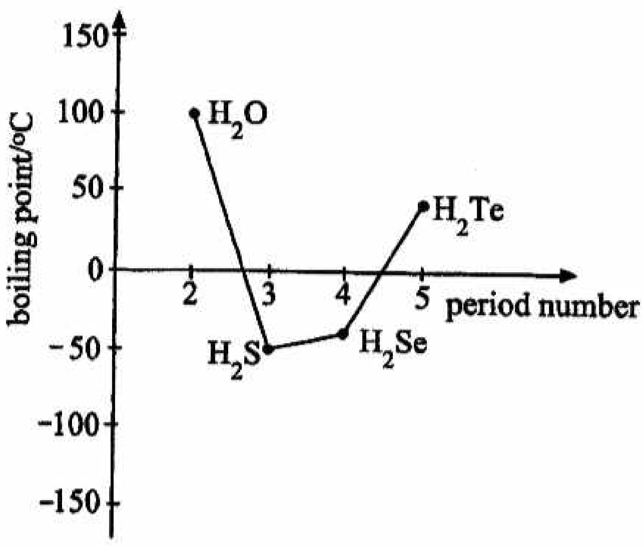

Fig. 2.2

Explain the trend in the boiling points of the hydrides. [5]

(c) (i) Show, with the aid of a diagram, the bonding in boron trifluoride.

(ii) Explain the reactivity of boron trifluoride.

(iii) Draw a diagram to illustrate the shape of the product formed when boron trifluoride reacts with ammonia. [3]

[Total: 12]

3 (a) The equation for the formation of ethanol is

$$2C_{(s)} + 3H_{2(g)} + \frac{1}{2}O_{2(g)} \rightarrow C_2H_5OH_{(\ell)}.$$

(i) Define the term *standard enthalpy change of formation*, ΔH^{θ}_{f}.

(ii) Calculate the ΔH^{θ}_{f} of ethanol given that:

$$\Delta H^{\theta}_{c}(C) = -393.5 \text{ KJ mol}^{-1}$$

$$\Delta H^{\theta}_{c}(H_2) = -285.8 \text{ KJ mol}^{-1}$$

$$\Delta H^{\theta}_{c}(C_2H_5OH) = -1\,371 \text{ KJ mol}^{-1}$$

(iii) Suggest why the ΔH^{θ}_{f} of ethanol cannot be measured directly.

[5]

(b) Explain why the ΔH^{θ}_{f} of H_2O is the same as the ΔH^{θ}_{c} of H_2. [2]

(c) In an experiment to measure the ΔH^{θ}_{c} of ethanol by calorimetry the following results were obtained:

volume of water in calorimeter	=	200.00 cm^3
mass of burner and ethanol at the beginning	=	81.00 g
mass of burner and ethanol at the end	=	77.78 g
initial temperature of water	=	24.0 °C
final temperature of water	=	89.0 °C

(i) Calculate the

1. heat energy absorbed by the water,

2. enthalpy change of combustion, ΔH^{θ}_{c}, of ethanol.
[Use of the data booklet is recommended]

(ii) Suggest why the ΔH^{θ}_{c} value of ethanol given in **a(ii)** is different from the experimental value. [5]

[Total:12]

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

For Performance Measurement

General Certificate of Education Advanced Level

CHEMISTRY

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9189/1

NOVEMBER 2015 SESSION

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[Turn over

Section A

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1 (a) **Fig.1.1** shows the trend in the boiling points of hydrides of Group (V) elements.

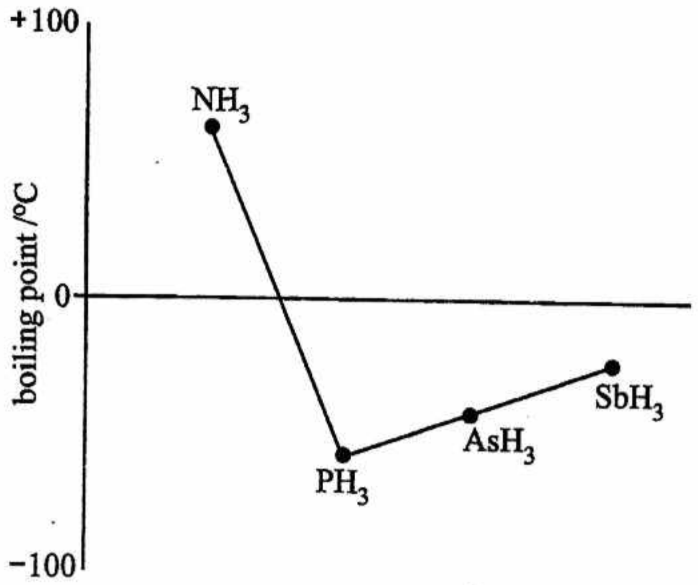

Fig.1.1

Explain the trend.

[2]

(b) Explain the following

(i) methanol is soluble in water but octanol is insoluble,

(ii) ethanoic acid has an apparent molar mass of $120\,gmol^{-1}$ in the vapour state just above its boiling point,

(iii) boron tri-chloride reacts with ammonia but carbon tetrachloride does not,

(iv) the bond angle in phosphorus tri-chloride is 107°,

(v) carbon dioxide is a gas whereas germanium (IV) oxide is a solid at room temperature.

[10]
[Total:12]

2 (a) (i) Define the terms

1. *activation energy,*

2. *catalyst.*

(ii) Explain why the activation energy for the synthesis of ammonia in the Haber process is high.

[3]

(b) The equation for the Haber process is

$$N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)} \quad \Delta H^{\theta} = -92\ kJmol^{-1}$$

and the activation energy, for the uncatalysed decomposition of ammonia is $+335\ kJmol^{-1}$.

(i) Sketch, on the same axes, labelled reaction pathway diagrams of the catalysed and uncatalysed reaction.

(ii) Determine the activation energy of the uncatalysed formation of ammonia.

[5]

(c) In an experiment, 120 mole of $H_{2(g)}$ were mixed with 40.0 mole of $N_{2(g)}$ and pressurised in a 2.0 dm^3 vessel until equilibrium was reached when there was 40 % conversion of the reactants to ammonia.

(i) Write the K_c expression for the formation of ammonia.

(ii) Calculate the K_c and state its units.

[4]
[Total:12]

3 (a) **[Use of the Data Booklet is recommended in this question.]**

The standard zinc - copper cell is as shown:

$$Zn_{(s)} / Zn^{2+}_{(aq)} // Cu^{2+}_{(aq)} / Cu_{(s)}$$

(i) Define the term *standard cell potential.*

(ii) Calculate the standard cell potential for this cell,

(iii) Write the overall equation for the cell reaction.

(iv) Explain the effect of adding zinc granules on the value of E^{θ} cell.

[4]

(b) Outline how zinc impurities are removed during the electro-purification of copper.

[2]

(c) A sample of impure zinc dust of mass 0.350 g was added to an excess of aqueous iron (III) sulphate. The iron (II) produced required 100 cm^3 of 0.0200 $moldm^{-3}$ potassium manganate (VII) for titration.

(i) Write the equation for the reactions which take place.

(ii) Calculate the percentage purity of the zinc dust.

[6]
[Total:12]

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CHEMISTRY 9189/1

PAPER 1

JUNE 2016 SESSION 2 hours

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Section A

Answer at least two questions from this section.

1 **(a)** **(i)** Define the term *ideal gas*.

(ii) Explain why noble gases behave more ideally than other gases.

[3]

(b) **(i)** A 185.00 g sample of fluorine gas and 4 moles of xenon were placed in a closed flask at 0 °C and 2.5 atm.

Deduce the volume of the gaseous mixture.

(ii) Addition of 23.00 g of lithium metal into the flask resulted in a violent reaction.

Write a chemical equation for the violent reaction.

(iii) Calculate the change in the

1. number of moles of fluorine,
2. total pressure.

[9]
[Total: 12]

2 **(a)** A Born-Haber Cycle for the formation of calcium sulphide is shown in **Fig. 2.1**.

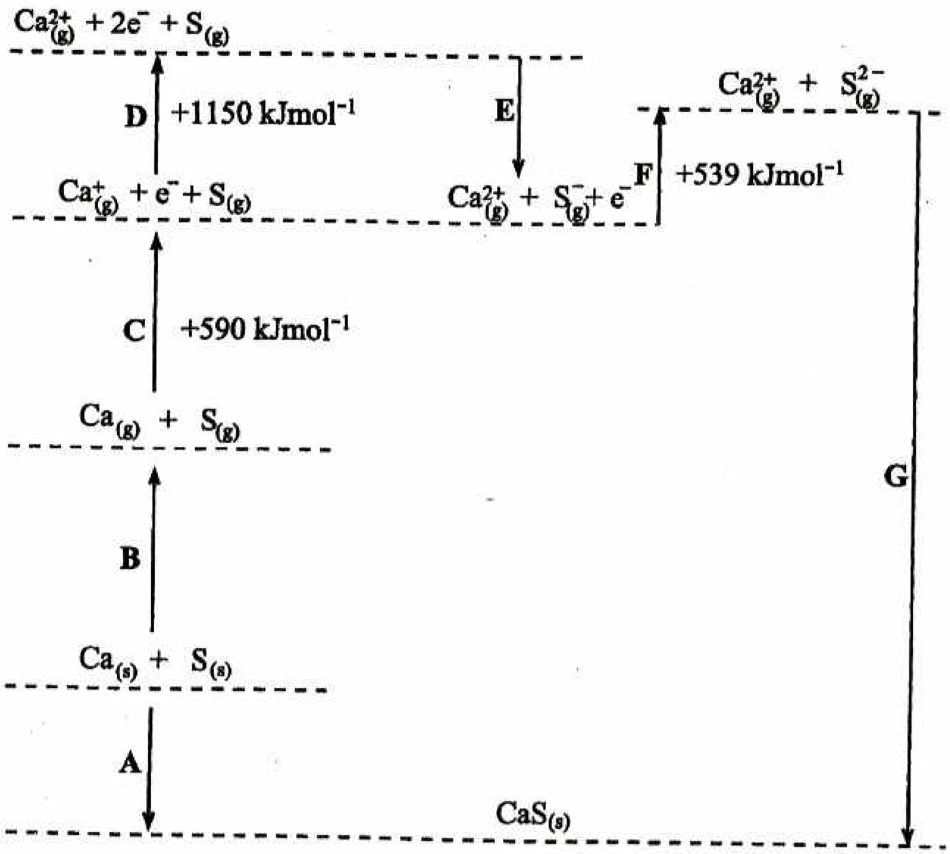


Fig. 2.1

(i) State the electronic configuration of the sulphide ion, S^{2-}.

(ii) Name enthalpy changes **D** and **E**.

(iii) Explain why the enthalpy change for **D** is larger than for **C**.

(iv) Suggest why **F** is an endothermic process.

[6]

(b) When a 0.218 mole sample of hydrogen iodide was heated in a flask of volume, V dm^3, the equilibrium established at 700 K was

$$2HI_{(g)} \rightleftharpoons H_{2(g)} + I_{2(g)}.$$

The equilibrium mixture was found to contain 0.023 moles of hydrogen.

(i) State and explain the effect of increasing pressure on the equilibrium.

(ii) Calculate the value of K_c at 700 K for the equilibrium.

[6]
[Total: 12]

[Turn over

3 **(a)** In the human body, one important buffer system in the blood involves the hydrogen carbonate ion, HCO_3^-, and carbonic acid, H_2CO_3. The equilibrium involved is

$$H_2CO_{3(aq)} \rightleftharpoons H^+_{(aq)} + HCO^-_{3(aq)}.$$

(i) Explain how the HCO_3^- / H_2CO_3 buffer system maintains a constant pH in the blood stream.

(ii) In a sample of blood with a pH of 7.41, the concentration of $HCO^-_{3(aq)}$ ions is 2.5×10^{-2} $moldm^{-3}$ and the concentration of $H_2CO_{3(aq)}$ is 1.2×10^{-3} $moldm^{-3}$.

Calculate the value of the acid dissociation constant for carbonic acid.

[4]

(b) A buffer solution made from mixing ethanoic acid and sodium ethanoate had 0.15 $moldm^{-3}$ ethanoic acid and 0.10 $moldm^{-3}$ sodium ethanoate. The K_a value for ethanoic acid is 1.74×10^{-5} $moldm^{-3}$ at 298 K.

Calculate the

(i) pH of the buffer solution,

(ii) change in pH on addition of a $10\ cm^3$ portion of 1.0 $moldm^{-3}$ hydrochloric acid to $1\ dm^3$ of the buffer solution.

[8]
[Total: 12]

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

CHEMISTRY

9189/1

PAPER 1

NOVEMBER 2016 SESSION

2 hours

Additional materials:
- Answer paper
- Data Booklet
- Mathematical tables/ electronic calculator
- Graph paper

TIME: 2 hours

INSTRUCTIONS TO CANDIDATES

Write your name, Centre number and candidate number in the spaces provided on the answer paper/answer booklet.

Answer **six** questions.

Answer **two** questions from Section A, **one** question from Section B, **two** questions from Section C and **one** other question chosen from any section.

Write your answers on the separate answer paper provided.

If you use more than one sheet of paper, fasten the sheets together.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [ ] at the end of each question or part question.

You are reminded of the need for good English and clear presentation in your answers.

This question paper consists of 9 printed pages and 3 blank pages.

Section A

*Answer at least **two** questions from this section.*

1 (a) An organic compound, A, was found to contain 70.6% C, 23.5% O and 5.9 % H.

(i) Define the term *empirical formula*.

(ii) Calculate the empirical formula of A. [3]

(b) The mass spectrum of A is shown in **Fig. 1.1**.

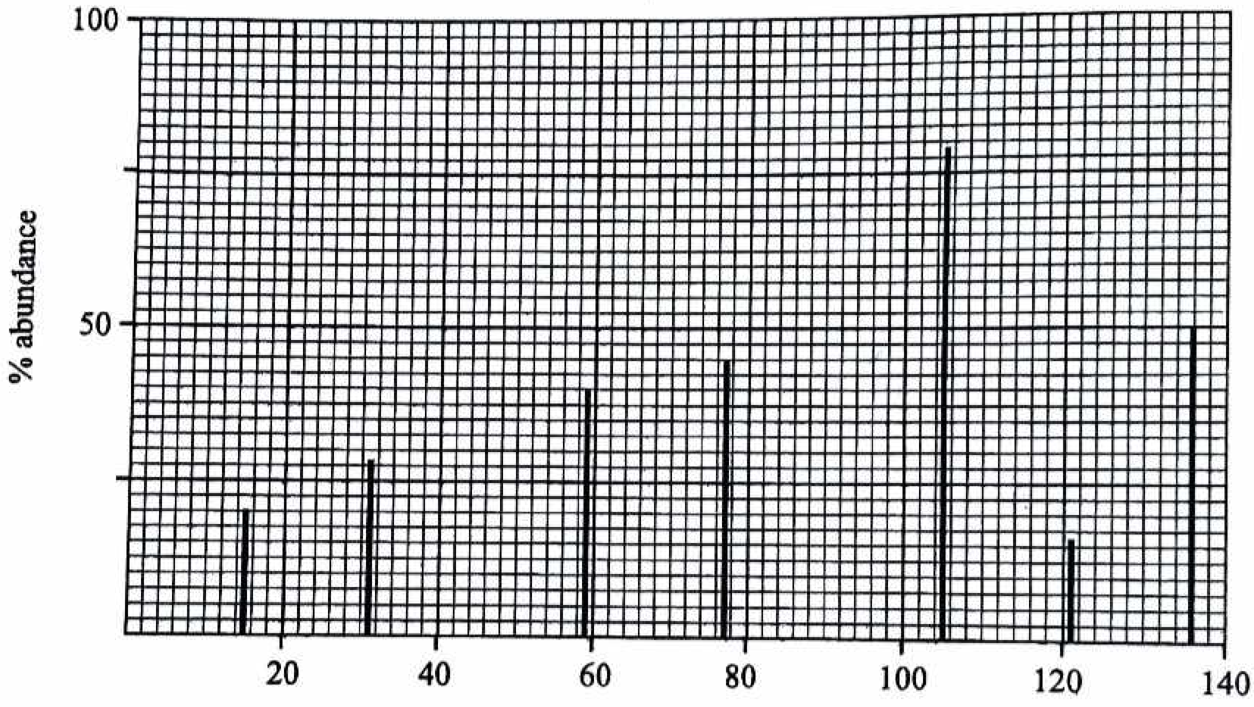


Fig. 1.1

(i) Deduce

1. the molecular formula of A.

2. the species giving rise to the peaks at m/e values of 31, 59 and 77.

(ii) Hence give the structural formula of A.

(iii) Calculate the relative molecular mass of a 1.00 g sample of A, given that its vapour occupied a volume of 218.20 cm^3 at 85°C and 1.01×10^5 Pa.

(iv) Compare the M_r value in **b(i)** and that in **b(iii)** and comment. [9]

[Total: 12]

2 (a) (i) Write equations to define the enthalpy changes of

1. formation of ethane,

2. atomisation of carbon.

(ii) Calculate the enthalpy change for the reaction

$$2C_{(g)} + 6H_{(g)} \rightarrow C_2H_{6(g)},$$

(iii) Calculate the C – C bond energy.

Given that

$$\Delta H_f^\theta(C_2H_6) = -88 \text{ kJmol}^{-1}$$
$$\Delta H_{at}^\theta(C_{(s)}) = +720 \text{ kJmol}^{-1}$$

[7]

(b) A temperature rise of 50 °C is produced when 10 cm^3 of concentrated sulphuric acid is added to 900 cm^3 of distilled water.

(i) Write a chemical equation for the change that causes the temperature rise.

(ii) The concentration of concentrated sulphuric acid is 18.0 $moldm^{-3}$ and the specific heat capacity of the resultant mixture is 4.2 $Jg^{-1} K^{-1}$.

Calculate

1. the heat change for the process,

2. ΔH_r^θ for the process. [5]

[Total: 12]

3 A 1.00 mole sample of ammonia was introduced into a 1.00 dm^3 vessel and allowed to dissociate at constant temperature. At equilibrium, it was 40% dissociated.

(a) **(i)** Write the equilibrium expression for the dissociation of ammonia.

(ii) Calculate the

1. equilibrium concentration of each component,

2. equilibrium constant, K_c.

[6]

(b) State and explain the effect on the equilibrium composition of

(i) increasing the volume of the container at constant temperature,

(ii) adding iron filings into the vessel.

[4]

(c) Sketch a graph to show how the concentration of ammonia changes with time. [2]

[Total: 12]

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

CHEMISTRY — 9189/1

PAPER 1

JUNE 2017 SESSION — 2 hours

Additional materials:
Answer paper
Data Booklet
Mathematical tables and/or electronic calculator

TIME: 2 hours

INSTRUCTIONS TO CANDIDATES

Write your name, Centre number and candidate number in the spaces provided on the answer paper/answer booklet.

Answer **six** questions.

Answer **two** questions from Section A, **one** question from Section B, **two** questions from Section C and **one** other question chosen from any section.

Write your answers on the separate answer paper provided.

If you use more than one sheet of paper, fasten the sheets together.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [ ] at the end of each question or part question.

You are reminded of the need for good English and clear presentation in your answers.

This question paper consists of 7 printed pages and 1 blank page.

Section A

*Answer at least **two** questions from this section.*

1 **(a)** **(i)** Define the term *equilibrium constant.*

(ii) State the factor that affects an equilibrium constant.

(iii) Explain how the factor in **(ii)** affects an equilibrium constant.

[3]

(b) Samples of 1.00 mol of Y and 0.75 mol of Z were mixed and allowed to reach equilibrium in a closed container.

They react according to the equation:

$$2Y_{(aq)} + Z_{(aq)} \rightleftharpoons X_{(aq)}$$

At equilibrium 0.70 mole of Y and 0.60 mol of Z were found to be present in the reaction mixture.

Calculate the equilibrium constant. [3]

(c) A titration was carried out using acidified potassium dichromate (VI) in a conical flask and iron (II) sulphate in a burette.

(i) Write a balanced equation for the reaction taking place.

(ii) Calculate the E^{θ}_{cell}.

(iii) Name the indicator used in this titration.

(iv) Sketch a graph to show how the e.m.f. changes during the titration.

[6]
[Total: 12]

2 **(a)** A gaseous hydrocarbon of volume 10 cm^3 was mixed with 50 cm^3 of excess oxygen and the mixture sparked so that the hydrocarbon was completely burnt. The volume of gas remaining at the end of combustion was 35 cm^3. After passing over sodalime, this volume was reduced to 5 cm^3. All gases were measured at 25 °C and at the same pressure.

(i) Calculate the volume of

1. oxygen used,

2. carbon dioxide produced.

(ii) Deduce the molecular formula of the hydrocarbon.

[5]

(b) **(i)** Define the term *standard enthalpy change of solution.*

(ii) Describe in terms of energetic processes the dissolution of an ionic compound.

(iii) Suggest the reasons for the low solubility of MgO in water.

[7]
[Total: 12]

3 **(a)** Hydrogen carbonate ions provide a buffering action in biological systems.

(i) Define the term *buffer solution.*

(ii) Explain by means of equations, the buffering action of hydrogen carbonate ions.

[3]

(b) A mixture of benzoic acid and sodium benzoate can also act as a buffer.

[K_a of benzoic acid is $6.4 \times 10^{-5} moldm^{-3}$]

Calculate the

(i) concentration of H^+ ions in 0.02 $moldm^{-3}$ benzoic acid,

(ii) pH of 0.02 $moldm^{-3}$ benzoic acid,

(iii) pH of a solution containing 7.2 g of sodium benzoate in 1 dm^3 of 0.02 $moldm^{-3}$ benzoic acid,

(iv) pH change, if 1 cm^3 of 1.0 $moldm^{-3}$ NaOH is added to the buffer in **(iii)**.

[7]

(c) Show that at 25 °C, the pH of water is 7. [2]

[Total: 12]

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

CHEMISTRY **6031/3**

PAPER 3

NOVEMBER 2018 SESSION 2 hours 30 minutes

Additional materials:
- Answer paper
- Data Booklet
- Mathematical tables and/or electronic calculator

TIME: 2 hours 30 minutes

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces provided on the answer paper/answer booklet.

Answer **six** questions.

Answer **two** questions from Section A, **one** question from Section B, **two** questions from [illegible] **and one** question from Section D.

Write your answers on the separate answer paper provided.

If you use more than one sheet of paper, fasten the sheets together.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [ ] at the end of each question or part question.

You are reminded of the need for good English and clear presentation in your answers.

Section A

Answer any two questions from this section.

(a) Fig.1.1 shows the mass spectrum of a sample of an element **Y**.

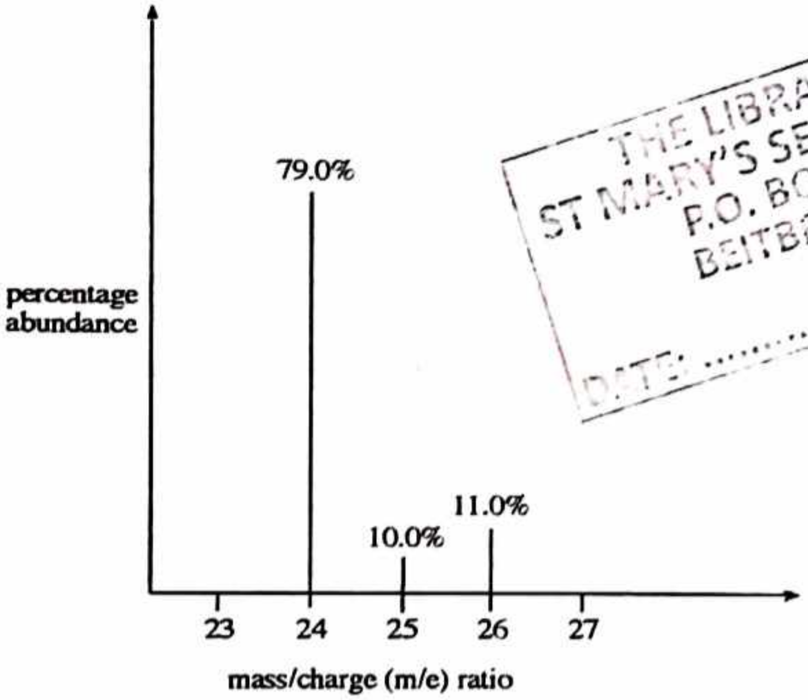


Fig.1.1

(i) Define the term *relative atomic mass.*

(ii) Calculate the relative atomic mass of Y.

(iii) Deduce the full electronic configuration of Y.

(iv) Identify the species responsible for the tallest peak.

(v) Explain why there are three different peaks in the mass spectrum of Y. [6]

(b) (i) Define the term *ionic product of water*, K_w.

(ii) Explain, using K_w, why at 25°C water has a pH of 7. [4]

(c) An aqueous buffer solution containing 0.055 moles of weak acid, HA, and 0.025 moles of NaA in 100 cm^3 of solution has a pH of 4.20.
[K_a of the acid, HA, is 2.87×10^{-5} $moldm^{-3}$ at 25°C]

Calculate the pH of the solution formed when 10 cm^3 of 0.130 $moldm^{-3}$ sodium hydroxide are added to 100 cm^3 of the buffer solution. [5]

[Total:15]

2 (a) **Table 2.1** shows enthalpy changes of some chemical processes.

Table 2.1

standard enthalpy of atomisation of magnesium	$+150\ kJmol^{-1}$
first electron affinity of oxygen	$-142\ kJmol^{-1}$
standard enthalpy of formation of magnesium oxide	$-602\ kJmol^{-1}$
lattice enthalpy of magnesium oxide	$-3\ 888\ kJmol^{-1}$

(i) Draw a Born-Haber cycle for the formation of magnesium oxide.

(ii) Calculate the second electron affinity of oxygen.

(iii) Explain why

1. the second electron affinity of oxygen is endothermic.

2. magnesium oxide is used as a refractory lining in furnaces.

[8]

(b) In an experiment to determine the rate of reaction between $S_2O_8^{2-}$ and I^- ions, in aqueous solution, the data in **Table 2.2** was obtained.

Table 2.2

experiment number	$[S_2O_8^{2-}]$/ $moldm^{-3}$	$[I^-]$ $moldm^{-3}$	initial rate/ $moldm^{-3}s^{-1}$
1	0.080	0.034	2.2×10^{-4}
2	0.080	0.017	1.1×10^{-4}
3	0.160	0.017	2.2×10^{-4}

(i) Write a balanced ionic equation for the reaction.

(ii) Determine the rate equation.

(iii) Calculate the rate constant, k.

(iv) Suggest an experimental method than can be used to determine the rate of this reaction.

[7]

[Total: 15]

3 (a) (i) Write the overall equation of the reaction in a hydrogen-oxygen fuel cell operating under alkaline conditions.

(ii) Calculate the e.m.f. of the cell.

(iii) State one advantage of fuel cells over lead-acid accumulators.

(iv) Explain why the e.m.f. of an acidic hydrogen-oxygen fuel cell is exactly the same as that of an alkaline fuel cell.

(v) State **one** advantage, other than cost, of using porous coated electrodes in fuel cells.
[use of data from the data booklet is recommended] [5]

(b) A hydrogen-oxygen fuel cell is required to produce a current of 0.03 A for 50 days.

Calculate the mass of hydrogen gas that will be needed. [3]

(c) Sulphur dioxide present in an air sample was dissolved in water. The solution was then titrated against acidified potassium manganate (VII) solution. 7.40 cm^3 of 3.16 gdm^{-3} potassium manganate (VII) solution was required to reach the endpoint.

(i) Construct a balanced chemical equation for the reaction.

(ii) Determine the mass of sulphur dioxide in the air sample.

[7]
[Total:15]

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

CHEMISTRY

PAPER 3

6031/3

JUNE 2019 SESSION

2 hours 30 minutes

Additional materials:
Answer paper
Data Booklet
Mathematical tables and/or electronic calculator

TIME: 2 hours 30 minutes

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces provided on the answer paper/answer booklet.

Answer **six** questions.

Answer **two** questions from Section A, **one** question from Section B, **two** questions from Section C and **one** question from section D.

Write your answers on the separate answer paper provided.

If you use more than one sheet of paper, fasten the sheets together.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [ ] at the end of each question or part question.

You are reminded of the need for good English and clear presentation in your answers.

Section A

Answer any two questions from this section.

1 (a) (i) State and explain the trend in the 1^{st} ionisation energies of elements in the same group.

(ii) Successive ionisation energies of three elements, **A**, **B** and **C**, are given in **Table 1.1**.

Table 1.1

	ionisation energies/$kJmol^{-1}$			
element	**1^{st}**	**2^{nd}**	**3^{rd}**	**4^{th}**
A	494	4 560	6 940	9 540
B	418	3 070	4 600	5 860
C	403	2 633	3 860	5 080

State and explain the Group of the Periodic Table to which the three elements are most likely to belong.

[4]

(b) (i) Explain why the first ionisation energy of Boron is lower than that of Berylium.

(ii) Write an equation that represents the second ionisation energy of Boron.

[3]

(c) (i) Draw an energy level diagram to show the electronic configuration of Boron.

(ii) Describe the shapes of the s and the p orbitals.

[3]

(d) The solubility product of iron (II) hydroxide is $x\ mol^3dm^{-9}$.

(i) Calculate the solubility of iron (II) hydroxide in terms of x.

(ii) Describe and explain the effect on the value in (i) of

1. increasing the temperature,

2. dissolving the iron (II) hydroxide in $NaOH_{(aq)}$.

[5]

[Total: 15]

2 (a) A hydrogen-oxygen fuel cell is shown in **Fig. 2.1**.

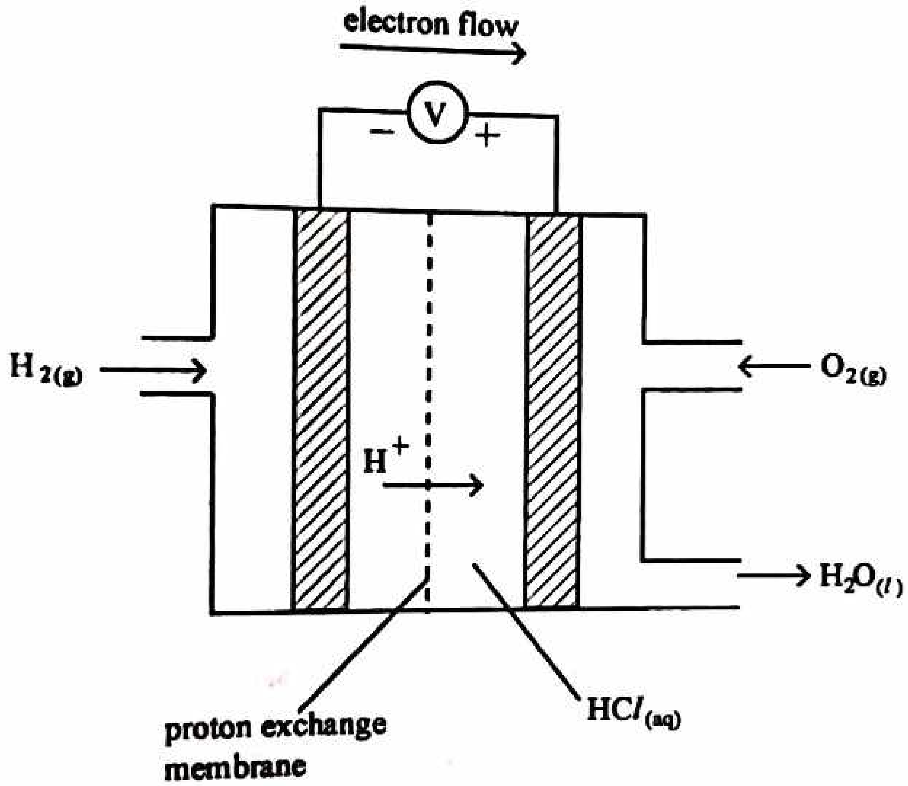


Fig. 2.1

(i) Write the equation for the overall cell reaction occurring in the hydrogen-oxygen fuel cell.

(ii) Calculate E^θ of the fuel cell.

[3]

(b) State any **two**

(i) advantages of using fuel cells compared to fossil fuels,

(ii) limitations to the use of fuel cells.

[3]

(c) An electrolytic cell was set up to determine the value of the Avogadro constant as shown in **Fig. 2.2**. 1.90 A of current were passed through the aqueous copper (II) sulphate for 50 minutes.

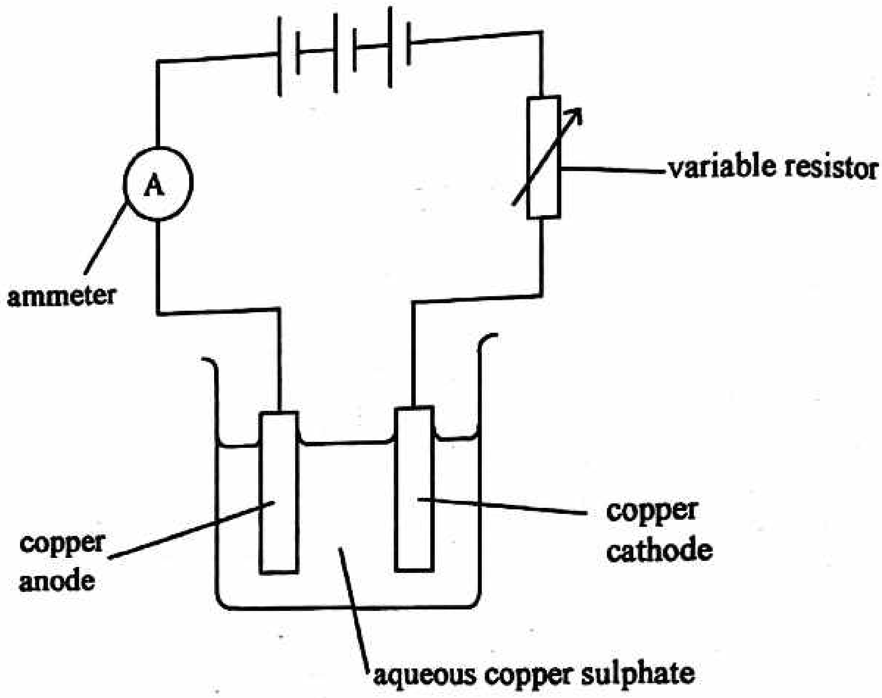


Fig. 2.2

(i) State any **three** observable changes that occurred.

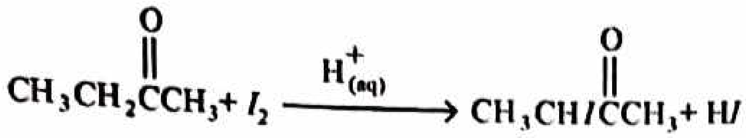
(ii) The mass of the cathode changed from 70.14 g to 71.97 g.

Calculate the Avogadro constant.

(iii) Explain why measuring the decrease in mass of the anode would be preferred to measuring the increase in mass of the cathode.

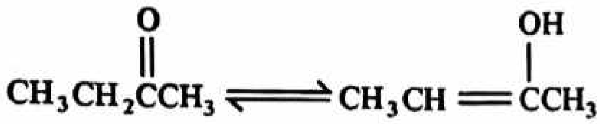
[9]
[Total 1

3 (a) Butanone reacts with iodine in an acid catalysed reaction as shown.



(i) Describe how the rate of this reaction can be determined.

(ii) An experimental investigation showed that the reaction is zero order with respect to iodine and that butanone exists in two interconvertible forms shown, in **Fig. 3.1**.



keto form *enol* form

Fig. 3.1

Propose a **two** step mechanism for the reaction.

[6]

(b) (i) Sketch graphs, on the same axes, to show how the rate of a reaction varies with concentration of a reactant for a

1. zero order,
2. first order and
3. second order reaction.

(ii) The overall order of a reaction was found to be third order.

Deduce the units of the rate constant for the reaction.

[4]

(c) Explain the following observations.

(i) Some collisions of particles do not lead to chemical reactions.

(ii) An increase in temperature increases the rate of a reaction.

[5]

[Total: 15]

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

CHEMISTRY **6031/3**

PAPER 3

NOVEMBER 2019 SESSION **2 hours 30 minutes**

Additional materials:
- Answer paper
- Data Booklet
- Electronic calculator

TIME: 2 hours 30 minutes

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces provided on the answer paper/answer booklet.

Answer six questions.

Answer two questions from Section A, **one** question from Section B, **two** questions from Section C and **one** question from Section D.

Write your answers on the separate answer paper provided.

If you use more than one sheet of paper, fasten the sheets together.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [ ] at the end of each question or part question.

You are reminded of the need for clear presentation in your answers.

This question paper consists of 13 printed pages and 3 blank pages.

Section A

Answer any two questions from this section.

1 (a) (i) Draw a dot and cross diagram to show the bonding in dichlorodifluoromethane.

(ii) Deduce, with reasons, the shape of dichlorodifluoromethane.

[3]

(b) An organic compound of mass 0.120 g was oxidised to 0.176 g carbon dioxide and 0.072 g water. When 0.148 g of the organic compound was vaporised at 60 °C, the vapour produced occupied a volume of 67.700 cm^3 at a pressure of 101 kPa.

(i) Determine the molecular formula of the organic compound.

(ii) Draw the displayed structural formula of the organic compound.

[7]

(c) An indigestion tablet of mass 0.50 g was dissolved in distilled water. The resultant solution required 28.30 cm^3 of 0.15 $moldm^{-3}$ of hydrochloric acid for complete neutralisation. The active ingredient in the tablet was found to be magnesium hydroxide.

(i) State the role of the indigestion tablet.

(ii) Determine the percentage by mass of the active ingredient in the tablet.

[5]

[Total:15]

2 (a) During respiration in humans, carbon dioxide produced diffuses into the blood to form a mixture of H_2CO_3 and HCO_3^-. The mixture acts as a buffer in the blood.

(i) Explain the importance of the H_2CO_3 and HCO_3^- mixture in the blood.

(ii) Write equations to show the buffering action of the mixture.

(iii) Calculate the mass of $NaHCO_3^-$ present in a 250 cm^3 sample of the buffer given that the pH of the buffer is 5.94 and the concentration of H_2CO_3 is 0.5 $moldm^{-3}$.
[$K_a(H_2CO_3) = 4.6 \times 10^{-7}$ $moldm^{-3}$]

[7]

(b) Ethanal and methanol react according to the equation:

$$CH_3CHO + 2CH_3OH \rightleftharpoons CH_3CH(OCH_3)_2 + H_2O$$

A 25.00 cm^3 mixture of the chemicals was found to contain 0.131 moles of CH_3CHO, 0.129 moles of CH_3OH, 0.0785 moles of $CH_3CH(OCH_3)_2$ and 0.0418 moles of H_2O.

(i) Calculate the K_c value, given that at equilibrium the amount of CH_3OH was found to be 0.0638 moles.

(ii) Describe and explain how the amount of $CH_3CH(OCH_3)_2$ can be increased.

[8]
[Total:15]

3 (a) **Fig. 3.1** shows the trend in the ionisation of magnesium.

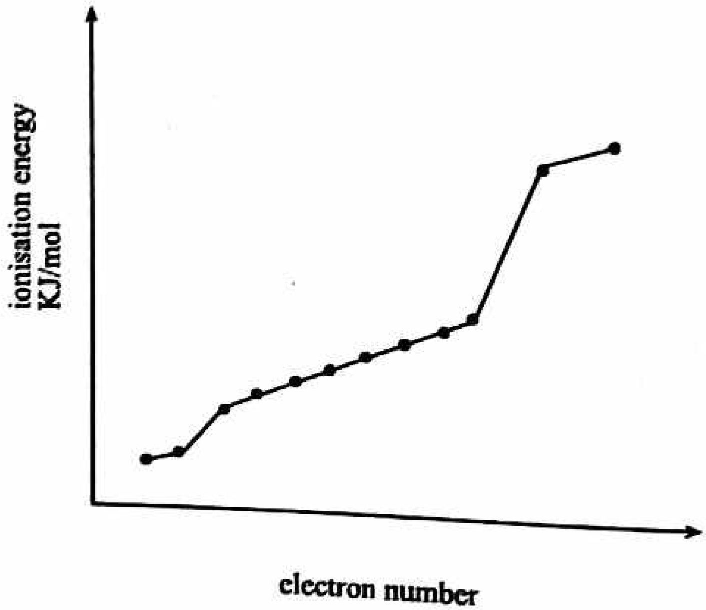


Fig. 3.1

(i) Define the term *first ionisation energy*.

(ii) Write an equation to illustrate the second ionisation of magnesium.

(iii) Explain how **Fig. 3.1** shows that there are three electron shells in magnesium.

[4]

(b) Aluminium exists as a mixture of three isotopes. Information about two of its isotopes is shown in the mass spectrum in Fig. 3.2.

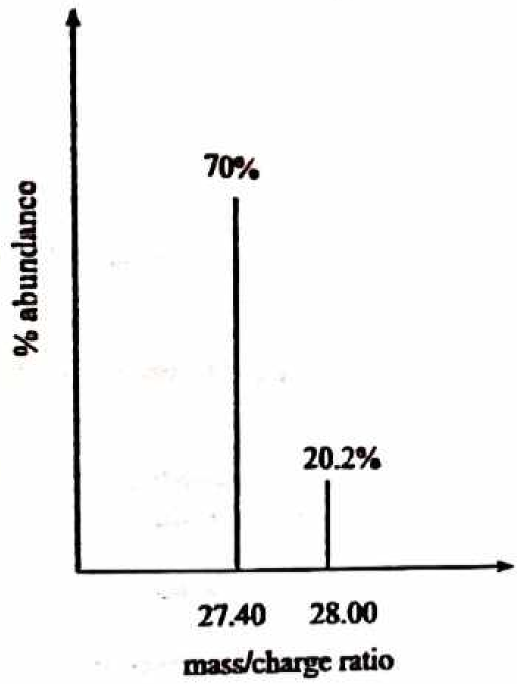


Fig. 3.2

Calculate the relative isotopic mass of the third isotope. [5]

(c) Explain why

(i) aluminium oxide is used as an insulator in spark plugs,

(ii) the first ionisation energy of potassium is lower than that of argon,

(iii) $SiCl_4$ is a liquid at room temperature whereas PCl_5 is a solid.

[6]
[Total: 15]

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

CHEMISTRY

6031/3

PAPER 3

JUNE 2020 SESSION **2 hours 30 minutes**

Additional materials
Answer paper
Data Booklet
Electronic calculator

TIME: 2 hours 30 minutes

INSTRUCTIONS TO CANDIDATES

Write your name, Centre number and candidate number in the spaces provided on the answer paper/answer booklet.

Answer **six** questions.

Answer **two** questions from Section A, **one** question from Section B, **two** questions from Section C and **one** question from section D.

Write your answers on the separate answer paper provided.

If you use more than one sheet of paper, fasten the sheets together.

INFORMATION FOR CANDIDATES

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You are reminded of the need for good English and clear presentation in your answers.

This question paper consists of 11 printed pages and 1 blank page.

[Turn over

Section A

Answer any ***two*** *questions from this section*

1 **(a)** An aldehyde, RCHO, reacts with acidified sodium cyanide according to the equation

$$RCHO + H^+ + CN^- \rightarrow RCH(OH)CN$$

Table 1.1 shows results of four experiments that were carried out to determine the kinetics of the reaction

Table 1.1

experiment number	[RCHO]/ moldm^{-3}	[H$^+$]/moldm^{-3}	[CN$^-$]/ moldm^{-3}	relative initial rate/moldm^{-3}s^{-1}
1	0.10	0.30	0.30	5.00
2	0.10	0.25	0.25	4.16
3	0.10	0.25	0.30	5.00
4	0.125	0.25	0.25	5.20

(i) Deduce the order of reaction with respect to

1. RCHO
2. H$^+$
3. CN$^-$

(ii) Calculate the value of the rate constant, K

[6]

(b) **(i)** Calculate the pH of 0.5 moldm^{-3} benzoic acid

[K_a of benzoic acid is 6.3×10^{-5} mol^{2}dm^{-6}]

(ii) Explain how a mixture of benzoic acid and sodium benzoate preserves food

[6]

(c) Describe and explain the observations made when iron filings are added to a solution of tin (IV) ions.

[3]

[Total: 15]

2 (a) Define the term

(i) *atomic radius*,

(ii) *shielding effect*. [2]

(b) Explain why

(i) the atomic radius of elements decrease across a period,

(ii) the cationic radius decrease from Na^+ to Al^{3+},

(iii) the anionic radius of an element is greater than the cationic radius,

(iv) the radius of the anion P^{3-} is greater than the radius of its non-metallic element P,

(v) there is an increase in atomic radius from Cl to Ar,

(vi) PCl_5 is a solid whereas PCl_3 is a liquid at r.t.p.

(vii) phosphonium ion, PH_4^+, is thermally unstable compared to the ammonium ion, NH_4^+. [7]

(c) $NOCl$ decomposes according to the equation.

$$2NOCl_{(g)} \rightleftharpoons 2NO_{(g)} + Cl_{2(g)}$$

A flask containing 0.50 moles of $NOCl$ was allowed to reach equilibrium at a pressure of 2.16 atmospheres. The equilibrium amount of $NOCl$ was found to be 0.30 moles.

Calculate the

(i) equilibrium amounts of

1. NO,

2. Cl_2,

(ii) equilibrium partial pressures of each of the three gases in the equilibrium mixture.

(iii) value of K_p.

[6]
[Total: 15]

3 **(a)** A mass of 2.00 g of an alloy of copper were dissolved in 50.00 cm^3 of concentrated nitric acid. The solution was made up to 200.00 cm^3 with distilled water. 20.00 cm^3 of this solution were reacted with aqueous potassium iodide. The iodine liberated reacted with 25.00 cm^3 of 1×10^{-1} $moldm^{-3}$ sodium thiosulphate.

Calculate the

(i) number moles of I_2 liberated by 20.00 cm^3 of $Cu^{2+}_{(aq)}$.

(ii) number moles of copper in 200.00 cm^3 of $Cu^{2+}_{(aq)}$.

(iii) percentage by mass of copper present in the alloy.

[7]

(b) **Table 3.1** shows values of enthalpies that describe the changes that occur when a solid $YX_{2(s)}$ dissolves in water

Table 3.1

enthalpy	value
lattice	$-2\ 327$ KJ mol^{-1}
hydration of Y^{2+}	$-1\ 920$ KJ mol^{-1}
hydration of X^-	-314 KJ mol^{-1}

(i) Define the term *enthalpy of hydration*

(ii) Calculate the enthalpy change of solution of $YX_{2(s)}$.

[3]

(c) Explain why plastic containers are used when preparing dilute sulphuric acid from the concentrated sulphuric acid in the laboratory.

[2]

(d) Carbonated drinks are preserved with $CO_{2(g)}$ dissolved under high pressure to produce the dynamic equilibrium:

$$CO_{2(g)} + H_2O_{(l)} \rightleftharpoons H_2CO_{3(aq)}$$

Explain why

(i) high pressure is necessary when sealing carbonated drinks.

(ii) rapid effervescence is observed when carbonated drinks are opened.

[3]
[Total: 15]

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

CHEMISTRY 6031/3

PAPER 3

NOVEMBER 2020 SESSION 2 hours 30 minutes

Additional materials:
- Answer paper
- Data Booklet
- Electronic calculator
- Graph paper

TIME: 2 hours 30 minutes

INSTRUCTIONS TO CANDIDATES

Write your name, Centre number and candidate number in the spaces provided on the answer paper/answer booklet.

Answer **six** questions.

Answer **two** questions from Section A, **one** question from Section B, **two** questions from Section C and **one** question from section D.

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INFORMATION FOR CANDIDATES

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You are reminded of the need for good English and clear presentation in your answers.

Section A

Answer any two questions from this section.

1 **(a)** Nitrosyl chloride, $NOCl_{(g)}$, dissociates into nitric oxide and chlorine when heated. A flask containing 0.50 moles of $NOCl_{(g)}$ was allowed to reach equilibrium at a pressure of 2.16 atm. The equilibrium amount of $NOCl_{(g)}$ was found to be 0.30 moles.

(i) Write a balanced chemical equation for the dissociation of $NOCl_{(g)}$.

(ii) Calculate the

1. equilibrium amounts of nitric oxide and chlorine,

2. K_p value. [8]

(b) **(i)** Write an equation to show how phenol reacts with water.

(ii) Calculate the concentration of phenol in a solution that was found to have a pH of 5.5.

[K_a of phenol = 1.6×10^{-10}moldm^{-3}]

(iii) Suggest why phenol is used in the manufacture of disinfectants.

[7]
[Total: 15]

2 **(a)** Define the term *bond energy*.

[1]

(b) Calculate the enthalpy change for the reaction of ethane with steam to produce ethanol.

[4]

[Use of relevant data from the data booklet is recommended]

(c) A volume of 25.00 cm^3 of 1.00 $moldm^{-3}$ NaOH, with an initial temperature of 20.0°C was added to 25.00 cm^3 of 1.00 $moldm^{-3}$ HCl. The initial temperature of the HCl was 21.0°C. The temperature of the mixture rose to 26.3 °C.

(i) Write an ionic equation for the reaction.

(ii) Calculate the

1. heat evolved,

2. value of the enthalpy change of neutralisation.

(Assume the heat capacity of the solution is 4.18 $Jg^{-1}k^{-1}$ and density of solution is $1gcm^{-3}$)

(iii) Explain why the calculated value in **(ii)2** differs from the theoretical value.

[10]
[Total: 15]

3 (a) A compound XY reacts with Y according to the equation:

$$XY + Y \rightarrow XY_2$$

The kinetics of the reaction was investigated using different concentrations of XY. The results are shown in **Table 3.1.**

Table 3.1

time/minutes	experiment I [XY] = 0.04 mol/dm^3	experiment II [XY] = 0.02 mol/dm^3
	[Y]	[Y]
0	0.01	0.01
15	0.008	0.009
30	0.0064	0.008
45	0.0052	0.0072
60	0.0042	0.0062
75	0.0034	0.0058
90	0.0028	0.0052

(II) Calculate the

1. initial rate of each of the two experiments,

2. order of reaction with respect to [Y].

(III) Deduce the order of reaction with respect to [XY].

[6]

(b) Explain the significance of reaction kinetics in industry. [4]

(c) Constant current electrolysis of dilute sulphuric acid at 0.75 A was carried out for 90 minutes under standard conditions.

Calculate the

(I) quantity of charge that passed through the circuit,

(II) volume of oxygen gas produced.

[5]
[Total: 15]

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

CHEMISTRY 6031/3

PAPER 3

NOVEMBER 2021 SESSION 2 hours 30 minutes

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INFORMATION FOR CANDIDATES

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This question paper consists of 14 printed pages and 2 blank pages.

[Turn over

Section A

Answer any two questions from this section.

1 (a) (i) Write the electronic configuration of Ca^{2+} ion.

(ii) Sketch a graph to show the variation in the metallic radii of Group (II) metals, Magnesium to Barium.

(iii) Explain the trend in the ease of formation of Group (II) cations.

[6]

(b) **Table 1.1** shows values of the lattice energies and solubilities of sulphates of Group (II) elements.

Table 1.1

compound	**lattice enthalpy/ $kJmol^{-1}$**	**solubility/moles per 100 g of water**
$MgSO_4$	−91.20	$3\ 600 \times 10^{-4}$
$CaSO_4$	+17.80	11×10^{-4}
$SrSO_4$	+18.70	062×10^{-4}
$BaSO_4$	+19.40	0.09×10^{-4}

(i) Draw an enthalpy diagram to show the energy changes that take place when magnesium sulphate dissolves in water.

(ii) Explain the reasons for the variation in the solubilities of the Group (II) sulphates.

(iii) Suggest a laboratory application for the low solubility of $BaSO_4$.

[5]

(c) Explain why

(i) calcium sulphate is a compound of medical importance,

(ii) calcium hydroxide is used in agriculture.

[4]
[Total: 15]

2 (a) Fig. 2.1 shows the melting point of Period 3 oxides, **A** to **G**.

[Letters **A** to **G** represent the oxides of the successive elements and are not chemical formulae.]

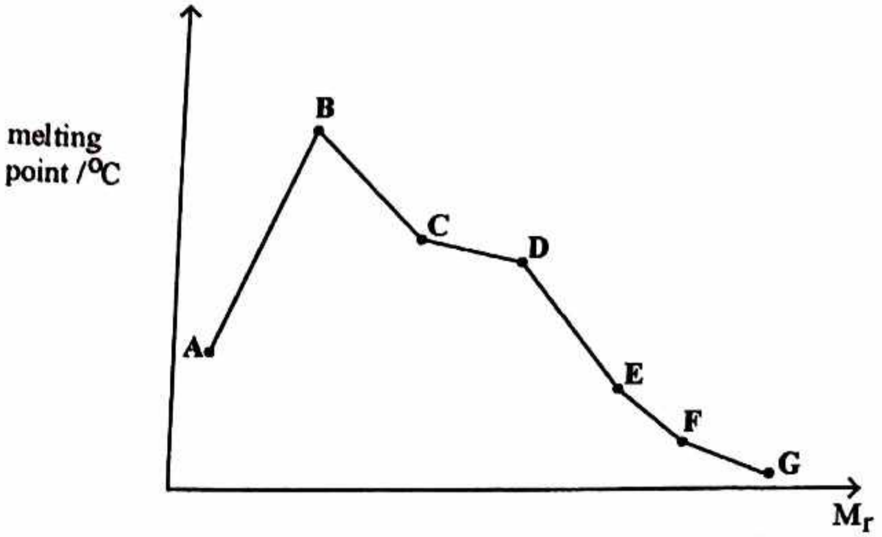


Fig. 2.1

(i) Name the lattice structure of **A**.

(ii) Explain why

1. the melting point rises sharply from **A** to **B**,

2. **D** has a higher melting point than **E**,

3. **C** only conducts electricity in molten state.

[5]

(b) Covalent bonds are formed between carbon atoms resulting in long chain and ring structures.

(i) State the importance of this carbon property,

(ii) Explain the carbon's ability to show this property.

[5]

[Turn over

(c) In an experiment, Pb_3O_4, was reacted with dilute nitric acid. The mixture was filtered and a dark brown solid separated.

(i) Write a balanced equation for the reaction that occured.

(ii) Explain how the reaction shows that PbO is a stronger base than PbO_2.

(iii) Galena, an impure ore of PbO contains 56% lead sulphide, PbS. 73 g of PbO was produced when 180 g of glena were heated in air.

$$2PbS + 2O_2 \rightarrow 2PbO + 2SO_2$$

Calculate

1. mass of lead sulphide in glena,

2. % yield of PbO.

(iv) Explain how the extraction of lead oxide from glena negatively affects the environment.

[5]

[Total: 15]

3 (a) State Le Chatelier's principle. [1]

(b) Gases A and B react according to the equation:

$$A_{2(g)} + B_{2(g)} \rightleftharpoons 2AB_{(g)}.$$

Table 3.1 shows the partial pressures obtained for the reaction at 500 °C.

Table 3.1

	partial pressure/Pa	
	initial	equilibrium
A	7.27×10^6	3.41×10^6
B	4.22×10^6	//////////////////////////
AB	0	7.72×10^6

(i) Deduce the partial pressure of B, at equilibrium.

(ii) Calculate the K_p for the reaction.

[3]

(c) Nitrogen reacts with hydrogen according to the equation

$$N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)} \; \Delta H_r = -92 \text{ kJmol}^{-1}.$$

(i) Explain how the reaction can be made to be as economic as possible for an industrial process.

(ii) State any **one** environmental problem that is associated with the use of nitrate fertilisers and explain its impact on the environment.

[6]

(d) **(i)** Define *common ion effect* as applied to ionic equilibria.

(ii) Explain why sodium chloride is added to the sodium stearate, $C_{17}H_{35}COONa$, solution during the manufacture of soap.

(iii) Calculate the solubility product of sodium stearate, given that the solubility of sodium stearate is 5×10^{-4} moldm^{-3}.

[5]
[Total: 15]

Section A

*Answer any **two** questions from this section.*

1 (a) (i) Draw diagrams to show orbital overlap for the bonding in

1. ethane,

2. ethene.

(ii) Differentiate between a sigma bond and pi bond.

[4]

(b) (i) State **three** conditions under which a gas approaches ideal gas behaviour. low temp

(ii) Identify, with reasons, which of the gases oxygen, hydrogen or nitrogen is

1. most ideal,

2. least ideal under the same temperatures and pressure conditions

(iii) A gaseous mixture of three gases contains 0.80 moles of gas A, 0.64 moles of gas B and 1.20 moles of gas C in a closed container at 25°C and 1 atmosphere.

Calculate the concentration of gas A in the mixture.

[11]

[Total: 15]

(a) **(i)** Explain the difference in the bond angles of NH_3 and CH_4.

(ii) Calculate the enthalpy change for the reaction shown, given that the N – O bond energy = 607 $kJmol^{-1}$.

$$2CO + 2NO \rightarrow 2CO_2 + N_2$$

[6]

[Use of relevant data from the data booklet is recommended]

(b) **(i)** Define the term

1. *weak acid*,

2. *buffer solution*.

(ii) Calculate the pH of a buffer solution prepared by adding $20.00\ cm^3$ of $0.40\ moldm^{-3}$ $HCl_{(aq)}$ to $30.00\ cm^3$ of $0.65\ moldm^{-3}$ $NH_{3(aq)}$, given that $K_b(NH_3) = 1.81 \times 10^{-5}\ moldm^{-3}$.

(iii) Give any **two** applications of buffers.

[9]

[Total: 15]

blood vessels
In Amino acids
In the formation of [illegible]

[Turn over

3 **(a)** Hydrogen peroxide decomposes slowly according to the equation.

$$2H_2O_{2(l)} \rightarrow 2H_2O_{(l)} + O_{2(g)}$$

A concentration time graph for the decomposition is shown in **Fig. 3.1**.

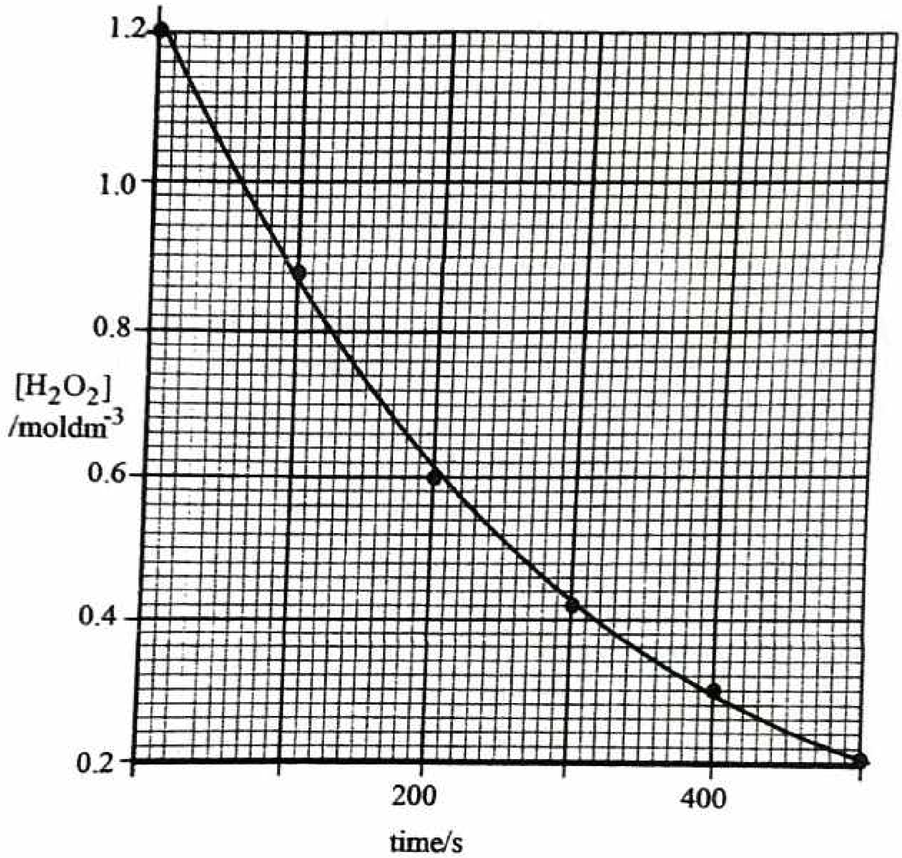


Fig. 3.1

(i) Deduce, using the graph, the

1. order of reaction,

2. rate of reaction at 200 seconds.

(ii) Explain how the rate of reaction at $t = 0$ seconds differs from that at $t = 200$ seconds.

(iii) Suggest **two** experimental methods that can be used to study the kinetics of the reaction.

[7]

(b) (i) Define the term

1. *solubility*,

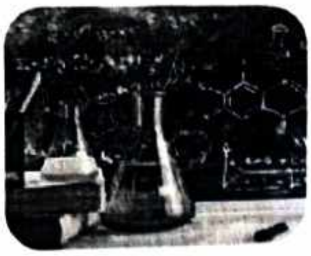
2. *ionic product*. [1]

(ii) Deduce whether a precipitate will form or not when 100 cm^3 of 0.001 $moldm^{-3}$ solution of magnesium chloride is mixed with 300 cm^3 of a 0.001 $moldm^{-3}$ sodium hydroxide solution.

[Ksp ($Mg(OH)_2$) = 7.16×10^{-11} mol^3dm^{-9}]

[8]

[Total: 15]



CHEMISTRY REVISION PAPER 3 OF 2023

TIME: 2 hours 30 minutes

INFORMATION FOR YOU.

A MUST READ!!!!

THE WRITER OF THIS PAPER HOLDS **NO** POSITION AT THE ZIMBABWE SCHOOL EXAMINATIONS COUNCIL OR ANY OTHER EXAMINATION BOARD THEREFORE THIS PAPER SHOULD **ONLY** BE VIEWED AS A SAMPLE PAPER AND SHOULD BE USED FOR PRACTICE ONLY.

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Section A

Answer any 2 questions from this section.

1 (a) (i) Define the terms:

1. solubility,
2. solubility product,
3. common ion effect.

(ii) Calcium phosphate, $Ca_3(PO_4)_2$, is a partially soluble salt with a solubility product of $5.81 \times 10^{-8}\ mol^5dm^{-15}$. Calculate the solubility of $Ca_3(PO_4)_2$ in:

1. water,
2. $0.3\ moldm^{-3}\ Na_3PO_4$.

Explain the difference in the answers obtained in 1 and 2 above.

[9]

(b) (i) calculate the pH of $0.25\ moldm^{-3}\ NH_3$.

$[K_a(NH_4^+) = 5.56 \times 10^{-10} moldm^{-3},\ K_w = 1.0 \times 10^{-14} moldm^{-3}]$.

(ii) Sketch a titration curve obtained when $0.25\ moldm^{-3}$ $50cm^3$ of NH_3 is titrated with $0.1\ moldm^{-3}\ HCl$.

(iii) Name the indicator appropriate for the titration above in (ii).

[6]

[TOTAL: 15]

2 (a) (i) Define the term *standard enthalpy change of formation.*

(ii) Write an equation the shows the enthalpy change of formation of Calcium Sulphide, CaS.

(iii) Calculate the enthalpy change of formation of calcium sulphide assuming that its combustion produces only CaO and SO_2.

ΔH_c	Energy/$kJmol^{-1}$
Ca	-635
CaS	-450
S	-297

[5]

(b) **Table 1.1** below shows enthalpy changes of some chemical reactions.

Table 1.1

Standard enthalpy of atomisation of Ca	+121 $kJmol^{-1}$
Standard atomisation of S	+279 $kJmol^{-1}$
First electron affinity of S	-200 $kJmol^{-1}$
Second electron affinity of S	+640 $kJmol^{-1}$

(i) Draw a born-haber cycle for the formation of CaS.

(ii) Using your answer from a(iii) above and some relevant data from the data booklet, calculate the lattice enthalpy of CaS.

(iii) Explain why the second electron affinity of sulphur is endothermic.

[6]

(c) Iron reacts with copper(II) according to the equation below:

$$Fe_{(s)} + Cu^{2+}_{(aq)} \longrightarrow Fe^{2+}_{(aq)} + Cu_{(s)}$$

(i) Write down the short hand description cell notation of the above reaction above.

(ii) Calculate the E^0_{cell} for the reaction.

(iii) State the effect, on the value of E^0_{cell}, of increasing the:

1. $[Fe^{2+}_{(aq)}]$.
2. thickness of the $Fe_{(s)}$ electrode.

[4]

[TOTAL: 15]

3 (a) A reacts reversibly with B according to the equation below:

$$2A_{(g)} + 3B_{(aq)} \longleftrightarrow 4C_{(g)} + 2D_{(s)} \quad \Delta H_r = -ve$$

(i) Write down an expression for the equilibrium constant K_c for the reaction.

(ii) State and explain the effect, on the position of equilibrium, of:
1. Decreasing the volume of the system,
2. Increasing temperature.

(iii) State the effect, on the value of the equilibrium constant K_c, of:
1. Increasing the concentration of B,
2. Increasing temperature.

(iv) State the effect of decreasing pressure on the equilibrium composition of C. [8]

(b) When 0.5 moles of A were allowed to react with 0.6 moles of B, the equilibrium moles of C was found to be 0.05.

Calculate the value of K_c including units given that the total volume is $2dm^3$. [7]

[TOTAL: 15]

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

CHEMISTRY 6031/3

PAPER 3

JUNE 2023 SESSION 2 hours 30 minutes

Additional materials:
Answer paper
Data Booklet
Electronic calculator

TIME: 2 hours 30 minutes

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces provided on the answer paper/answer booklet.

Answer **six** questions.

Answer **two** questions from Section A, **one** question from Section B, **two** questions from Section C and **one** question from Section D.

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INFORMATION FOR CANDIDATES

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You are reminded of the need for clear presentation in your answers.

This question paper consists of 12 printed pages.

Section A

Answer any two questions from this section.

1 (a) Manganese can be extracted from manganese (IV) oxide by reduction with carbon.

(i) Write an equation for the reduction process.

(ii) Calculate the enthalpy change for the extraction of manganese given that the enthalpy changes of formation of manganese (IV) oxide and carbon dioxide are $-520.3\ kJmol^{-1}$ and $-393.3\ kJmol^{-1}$ respectively.

(iii) Comment on the relative stability of manganese. [4]

(b) At 300 °C, gaseous phosphorous (V) chloride decomposes to give phosphorous (III) chloride gas, the process reaching an equilibrium.

(i) Write an equation for the decomposition process.

(ii) A sample of 0.12 moles of phosphorous (V) chloride in a closed vessel of 3 500 cm^3 was found to have 0.048 moles of phosphorous (III) chloride at equilibrium.

Calculate the

1 equilibrium concentrations of all components in the mixture,

2 K_c at this temperature. [6]

(c) The first six ionisation energies of an element **R** are shown in **Table 1.1**.

Table 1.1

ionisation number	1	2	3	4	5	6
ionisation energy / $kJmol^{-1}$	1000	1840	2880	4900	7820	16700

(i) Deduce, with reasons, the group of the periodic table to which element **R** belongs?

(ii) Give the electronic configuration of the outer most shell of **R**.

(iii) Draw diagrams to show the shapes of orbitals in the valency Shell of **R**.

[5]

[Total :15]

2 (a) When a 3.00 g sample of a mixture of $NaHCO_3$ and $NaCl$ was heated, the mass decreased by 0.31 g.

(i) Write a balanced chemical equation, for the reaction that occurred.

(ii) Calculate the mass of

1 $NaHCO_3$ in the sample,

2 $NaCl$ in the sample.

[5]

(b) At 30 °C, a solution of NaOH had a pH of 12.50.

(i) Calculate the concentration of NaOH given that K_w at 30 °C is 1.4×10^{-14} mol^2dm^{-6}.

(ii) Identify any **one** factor that affects K_w.

[5]

(c) Nickel is extracted from its ores by carbon reduction. The impure nickel produced is purified by electrolysis.

(i) Name any **one** metal element found in impure nickel.

(ii) Describe the electrolytic purification of the impure nickel.

[5]

[Total:15]

3 (a) The reaction between potassium manganate (VII), $KMnO_4$ and oxalic acid, $H_2C_2O_4$, is autocatalysed.

The oxidation half equation for the reaction is:

$$H_2C_2O_4 \rightleftharpoons 2CO_2 + 2e^- + 2H^+$$

(i) Name, with reasons, the type of reaction.

(ii) Define the term *autocatalysis*.

(iii) Sketch a graph to show how the rate of the reaction in (i) changes with time.

(iv) A 0.275 g sample of pure $H_2C_2O_4$ crystals were dissolved in dilute H_2SO_4, to which 0.2 g of pure $KMnO_4$ was added.

Calculate the volume of CO_2 produced at r.t.p.

[Use of relevant data from the data booklet is recommended] [7]

(b) The kinetics of a reaction involving three reactants, **A**, **B** and **C**, gave the results shown in **Table 3.1**.

Table 3.1

Experiment No.	Initial [A] / $moldm^{-3}$	Initial [B] / $moldm^{-3}$	Initial [C] / $moldm^{-3}$	Initial rate / $moldm^{-3}$
1	0.2	0.02	0.002	1.8×10^{-3}
2	0.2	0.04	0.002	3.6×10^{-3}
3	0.3	0.04	0.002	5.4×10^{-3}
4	0.3	0.02	0.004	2.7×10^{-3}

(i) Deduce the order of reaction with respect to the concentration of

1 **A**,

2 **B**,

3 **C**.

(ii) Calculate the rate constant.

(iii) Write the rate equation.

[8]

[Total:15]

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

CHEMISTRY — 6031/3
PAPER 3

NOVEMBER 2023 SESSION — 2 hours 30 minutes

Additional materials:
- Answer paper
- Data Booklet
- Electronic calculator

TIME: 2 hours 30 minutes

INSTRUCTIONS TO CANDIDATES

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This question paper consists of 10 printed pages and 2 blank pages.

[Turn over

Section A

Answer any ***two*** *questions from this section.*

1 **(a)** The isotopes of copper and their relative abundances are shown in **Table1.1**

Table1.1

isotope	height on graph (cm)
^{68}Cu	7
^{65}Cu	3

(i) Define the term *relative isotopic mass.*

(ii) Calculate the relative atomic mass, A_r, of copper. [3]

(b) **(i)** Define the term *lattice enthalpy.*

(ii) State and explain **two** factors that affect the magnitude of lattice enthalpy.

(iii) Explain why $\Delta H_{hyd}(Mg^{2+})$ is -1927 KJmol^{-1} whilst that of $Na^+ = -406$ KJmol^{-1}. [6]

(c) **(i)** State the **two** conditions for the formation of hydrogen bonding.

(ii) Sketch graphs of **Pv** against **P**, on the same axes, to show how the following gases deviate from ideal behaviour:

NH_3, CO_2 and He.

(iii) Explain the differences in the graphs. [6]

[Total :15]

2 (a) (i) Define the term *rate determining step*.

(ii) The reaction shown is second order with respect to NO and zero order with respect to CO and O_2.

$$NO_{(g)} + CO_{(g)} + O_{2(g)} \rightarrow NO_{2(g)} + CO_{2(g)}$$

Sketch curves to show how the rate of reaction changes with respect to

1. [NO],

2. [O_2].

(iii) At X °C the rate constant is 1.76×10^3 moldm^{-3} s^{-1} and at Y°C the rate constant is 5.8×10^3 moldm^{-3} s^{-1}.

Deduce, with a reason, whether Y is a higher or a lower temperature. [4]

(b) (i) Write an equation to show how H_3PO_4 reacts with water.

(ii) Identify conjugate acid-base pairs in the equation in **(i)**.

(iii) Predict shape and bond angles of the two product species in equation **(i)**. [4]

(c) During electrolysis of dilute Na_2SO_4 connected in series with $AgNO_{3(aq)}$ a current of 2 A was passed and 25 g of Ag was deposited.

(i) Calculate the time required to deposit 25 g of Ag.

(ii) Calculate the volume of gas produced at the anode in the Na_2SO_4 cell at r.t.p. [6]

(d) Explain why lead acid accumulators become flat after a period of discharge. [1]

[Total:15]

[1]

3 **(a)** State Le Chatelier's principle.

(b) Water exists in equilibrium with its ions as shown.

$$H_2O_{(l)} \rightleftharpoons OH^-_{(aq)} + H^+_{(aq)} \quad \Delta H^o = +58\ KJmol^{-1}$$

(i) State and explain the effect, on the position of the equilibrium, of

1. increasing the temperature of water,

2. adding HCl.

(ii) Write an expression for the equilibrium constant, K_w, and state its unit.

(iii) Explain why the value of K_w is affected by changes in temperature and **not** changes in concentration. [10]

(c) When iodic acid, HIO_3 is dissolved in water, it ionises slightly according to the following equilibrium:

$$HIO_{3(aq)} \rightleftharpoons H^+_{(aq)} + IO^-_{3(aq)}$$

In a 0.50 moldm^{-3} solution of HIO_3 the concentration of H^+ ions at equilibrium was found to be 0.22 moldm^{-3}.

Calculate the

(i) equilibrium concentration of $HIO_{3(aq)}$,

(ii) K_a value of HIO_3. [4]

[Total:15]